Chapter 3. Probability and Probability Distributions

Probability, random variables, expectation, covariance

Contents

3.1 Definition of probability **2**

3.2 Conditional probability and independence **3**

3.3 Random variables and probability distributions **5**

3.4 Expected value and its properties **7**

3.5 Joint distribution of two random variables **10**

3.6 Covariance and correlation **13**

3.1 Definition of probability

Motivation. Probability makes inductive conclusions from samples quantifiable; it is assigned to events.

Definition (Sample space and events)

The set of all possible outcomes of a statistical survey (experiment) is the **sample space** S. A subset of S—the set of outcomes satisfying some condition of interest—is an **event**; the event A *occurs* if the observed outcome is an element of A. An event consisting of a single element is an **elementary** (**simple**) **event**.
$A \cup B$, $A \cap B$, A^c are the **union**, **intersection** and **complement**. If $A \cap B = \varnothing$, A and B are **mutually disjoint** (exclusive): they cannot occur together.

Illustration 3-1: sample spaces

(a) Number shown by one die: $S = \{1, 2, 3, 4, 5, 6\}$.
(b) Number of items produced until the 10th defective: $S = \{10, 11, 12, \dots\}$.
(c) Lifetime of a light bulb drawn from a factory: $S = \{t \mid t \geq 0\}$.

For finitely many equally likely elementary events (Laplace):

Definition 3-1 (Classical definition of probability)

If $S = \{e_1, \dots, e_N\}$ has N elements whose elementary events are equally likely, the **probability** of an event A with m elements is

$$P(A) = \frac{m}{N}.$$

Example 3-1 (Table-tennis balls)

A box contains 3 white balls and 1 yellow ball. Two balls are drawn by simple random sampling. Find the probability of getting one white and one yellow ball.
Solution. Label the white balls a, b, c and the yellow one d:

$$S = \{\{a,b\},\{a,c\},\{a,d\},\{b,c\},\{b,d\},\{c,d\}\}, \qquad A = \{\{a,d\},\{b,d\},\{c,d\}\}.$$

All 6 pairs are equally likely, so $P(A) = 3/6 = 1/2$.

This fails for infinitely many or unequally likely outcomes. Under repetition the *relative frequency* of an event settles at a constant (Figure 3-1: proportion of defectives); Kolmogorov's axioms abstract its properties.

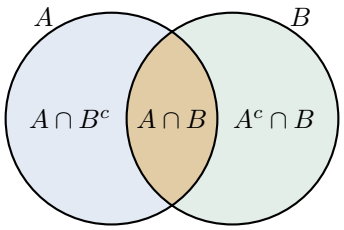
Definition 3-2 (Axiomatic definition of probability)

A rule P assigning numbers to events of S is a probability if
(1) $0 \leq P(A) \leq 1$ for every event A;
(2) $P(S) = 1$;
(3) for mutually disjoint events $A_1, A_2, \dots,$

$$P(A_1 \cup A_2 \cup \cdots) = P(A_1) + P(A_2) + \cdots.$$

$P(A)$ is called the **probability** of A.

This covers empirical (limiting frequency) and subjective probability.



Partition of $A \cup B$ into three disjoint pieces (Figure 3-2)

Theorem 3-1 (Properties of probability)

(1) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
(2) $P(A^c) = 1 - P(A)$.
(3) If $A \subset B$, then $P(A) \leq P(B)$.

Proof. (1) $A \cap B^c$, $A \cap B$, $A^c \cap B$ are disjoint with union $A \cup B$; by axiom (3),

$$P(A \cup B) = P(A \cap B^c) + P(A \cap B) + P(A^c \cap B),$$
$$P(A) = P(A \cap B^c) + P(A \cap B),$$
$$P(B) = P(A \cap B) + P(A^c \cap B).$$

Subtracting, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

(2) $P(A) + P(A^c) = P(S) = 1$ (axioms (3), (2)).

(3) $B = A \sqcup (A^c \cap B)$, so $P(B) = P(A) + P(A^c \cap B) \geq P(A)$ (axioms (3), (1)). ■

Example 3-2 (Three bags)

Each of three bags contains 5 marbles numbered 1 to 5. One marble is drawn from each bag. Find the probability that the sum of the numbers is at least 5.
Solution. $A =$ "sum ≥ 5" has $A^c = B \sqcup C$, $B =$ "sum $= 3$", $C =$ "sum $= 4$"; of $5^3 = 125$ outcomes,

$$B = \{(1,1,1)\}, \qquad C = \{(1,1,2), (1,2,1), (2,1,1)\},$$

so $P(B) = 1/125$, $P(C) = 3/125$ and

$$P(A) = 1 - \{P(B) + P(C)\} = 1 - \frac{4}{125} = \frac{121}{125}.$$

3.2 Conditional probability and independence

Motivation. Conditional probability updates for information; independence means no update is needed.

Given A, A acts as the sample space, and B gets the share $P(A \cap B)/P(A)$.

Definition 3-3 (Conditional probability)

The **conditional probability** of B given A is

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} \qquad (P(A) > 0).$$

Example 3-3 (Two dice)

Two dice are thrown. Given that the first die shows more than the second, find the probability that the sum is 7.

Solution. Let $A =$ "first $>$ second", $B =$ "sum $= 7$". Of 36 outcomes, A has 15, so $P(A) = 15/36$. Also $B = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$ and $A \cap B = \{(4,3),(5,2),(6,1)\}$, so

$$P(B) = \frac{6}{36} = \frac{1}{6}, \qquad P(A \cap B) = \frac{3}{36} = \frac{1}{12}, \qquad P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{3}{15} = \frac{1}{5}.$$

Theorem 3-2 (Multiplication rule)

If $P(A) > 0$ and $P(B) > 0$, then

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A).$$

Proof. Multiply $P(B \mid A) = P(A \cap B)/P(A)$ (Definition 3-3) by $P(A)$; exchange A and B. ∎

Example 3-4 (Prize draw)

A couple joins a draw at a supermarket: a box holds 10 winning and 50 losing balls; each person draws once and a drawn ball is not returned. Find the probability that both win.

Solution. Let $A =$ "husband wins", $B =$ "wife wins". Then

$$P(A \cap B) = P(B \mid A)P(A) = \frac{9}{59} \times \frac{10}{60} = \frac{3}{118}.$$

If $A_1, \dots, A_n$ *partition* S, the $B \cap A_i$ partition B.

Theorem 3-3 (Law of total probability)

If $A_i \cap A_j = \varnothing$ $(i \neq j)$, $A_1 \cup A_2 \cup \cdots \cup A_n = S$ and $P(A_i) > 0$, then

$$P(B) = P(B \mid A_1)P(A_1) + \cdots + P(B \mid A_n)P(A_n).$$

Proof.

Step 1. For $i \neq j$, $(B \cap A_i) \cap (B \cap A_j) \subset A_i \cap A_j = \varnothing$, and

$$(B \cap A_1) \cup \cdots \cup (B \cap A_n) = B \cap (A_1 \cup \cdots \cup A_n) = B \cap S = B.$$

Step 2. By axiom (3), $P(B) = \sum_{i=1}^{n} P(B \cap A_i)$, and by Theorem 3-2 each term equals $P(B \mid A_i)P(A_i)$. ∎

Example 3-5 (Joint training)

Three football teams train together: 50% of the players are from team 1, 30% from team 2, 20% from team 3; 20%, 15%, 30% of the respective teams are forwards. A player is chosen at random. Find the probability that he is a forward.
Solution. With A_i = "from team i" (a partition) and B = "forward",

$$P(B) = 0.2 \times 0.5 + 0.15 \times 0.3 + 0.3 \times 0.2 = 0.205.$$

If $P(B \mid A) = P(B)$, then $P(A \cap B) = P(A)P(B)$ and, by the multiplication rule, $P(A \mid B) = P(A)$.

Definition 3-4 (Independent events)

Events A and B are **independent** if one of the following holds (when the conditional probabilities are defined, they are equivalent):

$$P(A \cap B) = P(A)P(B), \qquad P(A \mid B) = P(A), \qquad P(B \mid A) = P(B).$$

Example 3-6 (Prize draw revisited)

In Example 3-4:
(a) Are "husband wins" and "wife wins" independent?
(b) Are they independent if the drawn ball is returned to the box?
Solution.
(a) By the law of total probability,

$$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c) = \frac{9}{59} \cdot \frac{10}{60} + \frac{10}{59} \cdot \frac{50}{60} = \frac{1}{6}.$$

Since $P(B) \neq P(B \mid A) = 9/59$, A and B are not independent.
(b) With replacement, $P(B \mid A) = 10/60 = 1/6 = P(B)$, so A and B are independent.

Draws with replacement are independent, without replacement not. Non-independent events are **dependent**. In general, $A_1, \ldots, A_n$ are **independent** if

$$\begin{aligned} P(A_i \cap A_j) &= P(A_i)P(A_j) && (1 \le i < j \le n), \\ P(A_i \cap A_j \cap A_k) &= P(A_i)P(A_j)P(A_k) && (1 \le i < j < k \le n), \\ &\vdots \\ P(A_1 \cap A_2 \cap \cdots \cap A_n) &= P(A_1)P(A_2)\cdots P(A_n). \end{aligned}$$

3.3 Random variables and probability distributions

Motivation. Mapping outcomes to numbers moves probability from the sample space to sets of numbers.

Illustration 3-2: coin toss

For $S = \{\mathrm{H}, \mathrm{T}\}$, $X(\mathrm{H}) = 1$, $X(\mathrm{T}) = 0$ transfers the probability on S to $\{0, 1\}$: $P(X = 0) = P(X = 1) = 1/2$.

Definition (Random variable, distribution)

A real-valued function defined on a sample space is a **random variable**. If an outcome is mapped to the value a, we say "X takes the value a". The description of how probability is spread over the values of X, by nonnegative numbers summing to 1, is the **probability distribution** (or simply **distribution**) of X.

Illustration 3-3

In Illustration 3-2 the distribution of X is $P(X=0)=P(X=1)=1/2$.

If the possible values $x_1, x_2, \dots$ of X can be listed (countable), X is a **discrete random variable**, and

$$p(x)=\begin{cases} P(X=x_i), & x=x_i \ (i=1,2,\dots) \\ 0, & \text{otherwise} \end{cases}$$

is its **probability mass function**.

Definition 3-5 (Probability mass function of a discrete r.v.)

A function $p(x)$ is the **probability mass function** of the discrete random variable X if
(1) $0 \le p(x) \le 1$ and $\sum_{\text{all } x} p(x) = 1$;
(2) $P(a < X \le b) = \sum_{a<x\le b} p(x)$.

The p.m.f. determines the distribution; with few values, list it as a table.

Example 3-7 (Popular posts)

Of an influencer's 12 posts, 4 have more than 10,000 likes. Three posts are chosen by simple random sampling; X is the number among them with more than 10,000 likes. Find the distribution of X.
Solution. There are $\binom{12}{3}=220$ equally likely choices, and $p(x)=\binom{8}{3-x}\binom{4}{x}/\binom{12}{3}$:

$$p(0)=\frac{56\times 1}{220}=\frac{14}{55}, \qquad p(1)=\frac{28\times 4}{220}=\frac{28}{55},$$
$$p(2)=\frac{8\times 6}{220}=\frac{12}{55}, \qquad p(3)=\frac{1\times 4}{220}=\frac{1}{55}.$$

If X can take every value in an interval, it is a **continuous random variable**.

Illustration 3-4: spinner

A pointer on a dial marked 0 to 1 spins freely; $X \in [0,1]$ is where it stops.

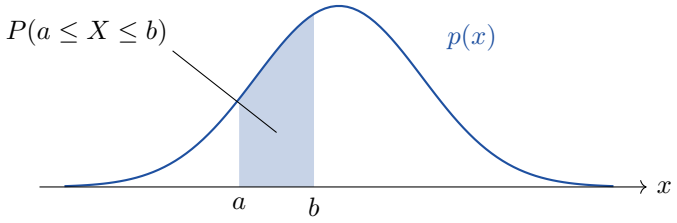
By the ratio of lengths,

$$P(a \le X \le b)=\frac{b-a}{1-0}=b-a \qquad (0 \le a < b \le 1).$$

With $p=1$ on $[0,1]$ and 0 otherwise, $P(a \le X \le b)=\int_a^b p(x)\,dx$: p is the **probability density function**.

Definition 3-6 (Probability density function of a continuous r.v.)

A function $p(x)$ is the **probability density function** of the continuous random variable X if
(1) $p(x) \geq 0$ and $\int_{-\infty}^{\infty} p(x)\,dx = 1$;
(2) $P(a \leq X \leq b) = \int_a^b p(x)\,dx$ $(-\infty \leq a < b \leq \infty)$.



$P(a \leq X \leq b)$ is the area under the density over $[a, b]$ (Figure 3-5)

Theorem 3-4 (Properties of continuous random variables)

For a continuous random variable X,
(1) $P(X = c) = 0$ for every c;
(2) $P(a \leq X \leq b) = P(a \leq X < b) = P(a < X \leq b) = P(a < X < b)$.

Proof. (1) $\{X = c\} \subset \{c - h \leq X \leq c + h\}$, so by Theorem 3-1(3)

$$0 \leq P(X = c) \leq \int_{c-h}^{c+h} p(x)\,dx \longrightarrow 0 \qquad (h \to 0).$$

(2) The sets differ by $\{X = a\}$, $\{X = b\}$, of probability 0 by (1); apply axiom (3). ■

Example 3-8 (Finding the constant)

A continuous random variable X has density $p(x) = bx(1 - x)$ for $0 \leq x \leq 1$, and 0 for $x < 0$ or $x > 1$. Find b and $P(0 \leq X \leq \frac{2}{3})$.
Solution. $1 = \int_0^1 bx(1 - x)\,dx = b/6$, so $b = 6$. Then

$$P\Big(0 \leq X \leq \frac{2}{3}\Big) = \int_0^{2/3} 6x(1 - x)\,dx = \Big[3x^2 - 2x^3\Big]_0^{2/3} = \frac{4}{3} - \frac{16}{27} = \frac{20}{27}.$$

An observed value of a population characteristic is a random variable with the population distribution; populations are modelled by such random variables.

3.4 Expected value and its properties

Motivation. The mean and variance of Chapter 2, for random variables.

Definition 3-7 (Expected value of a random variable)

If the p.m.f. (p.d.f.) of the discrete (continuous) random variable X is $p(x)$,

$$E(X) = \begin{cases} \sum_{\text{all } x} x\,p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} x\,p(x)\,dx & \text{(continuous)} \end{cases}$$

Example 3-9 (A mean)

X has distribution $p(1) = \frac{1}{8}$, $p(2) = \frac{3}{8}$, $p(3) = \frac{3}{8}$, $p(4) = \frac{1}{8}$. Find $E(X)$.
Solution. $E(X) = 1 \cdot \frac{1}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{3}{8} + 4 \cdot \frac{1}{8} = 2.5$, the centre of the distribution graph (Figure 3-6).

$E(X)$ is the centre of mass of masses $p(x_i)$ at x_i (or density p); for a population model it is the population mean μ.

Example 3-10 (A function of X)

A coin is tossed 3 times; X is the number of heads.
(a) Find the distribution of $Y = X(3 - X)$.
(b) Find $E(Y)$.
Solution.
(a) $p_X(0) = \frac{1}{8}$, $p_X(1) = \frac{3}{8}$, $p_X(2) = \frac{3}{8}$, $p_X(3) = \frac{1}{8}$. $Y = 0$ when $X \in \{0, 3\}$ and $Y = 2$ when $X \in \{1, 2\}$, so $p_Y(0) = \frac{1}{4}$, $p_Y(2) = \frac{3}{4}$.
(b) $E[X(3 - X)] = E(Y) = 0 \cdot \frac{1}{4} + 2 \cdot \frac{3}{4} = \frac{3}{2}$.

$E(Y)$ can be computed without the distribution of Y:

$$\begin{aligned} E[X(3-X)] &= 0(3-0)p_X(0) + 1(3-1)p_X(1) + 2(3-2)p_X(2) + 3(3-3)p_X(3) \\ &= 0 \cdot \tfrac{1}{8} + 2 \cdot \tfrac{3}{8} + 2 \cdot \tfrac{3}{8} + 0 \cdot \tfrac{1}{8} = \tfrac{3}{2}. \end{aligned}$$

Theorem 3-5 (Expectation of a function of a random variable)

If the p.m.f. (p.d.f.) of the discrete (continuous) random variable X is $p(x)$, then for any function g

$$E[g(X)] = \begin{cases} \sum_{\text{all } x} g(x)\,p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} g(x)\,p(x)\,dx & \text{(continuous)} \end{cases}$$

Proof. [Supplement]

Step 1 (discrete case). $\{Y = y\} = \bigsqcup_{x:\,g(x)=y}\{X = x\}$, so $p_Y(y) = \sum_{x:\,g(x)=y} p(x)$ and

$$E(Y) = \sum_y y\,p_Y(y) = \sum_y \sum_{x:\,g(x)=y} g(x)\,p(x) = \sum_{\text{all } x} g(x)\,p(x),$$

each x in exactly one group (regrouping valid under absolute convergence, the condition for $E(Y)$ to exist).

Step 2 (continuous case, monotone g). For differentiable strictly increasing g with inverse h, $P(Y \le y) = \int_{-\infty}^{h(y)} p(x)\,dx$, so $p_Y(y) = p(h(y))h'(y)$ and, with $x = h(y)$,

$$E(Y) = \int y\,p(h(y))h'(y)\,dy = \int g(x)\,p(x)\,dx.$$

For decreasing g, $p_Y(y) = -p(h(y))h'(y)$ and the limits reverse; same result.

Step 3 (general g). Split into intervals of monotonicity and apply Step 2; for arbitrary g this is the measure-theoretic change of variables, which we accept. ■

Example 3-11 (Waiting for a bus)

Buses leave a stop at 0, 15, 30 and 45 minutes past each hour. A person arrives at a random time. Find the expected waiting time.

Solution. The arrival time X (minutes past the hour) has density $\frac{1}{60}$ on $(0, 60]$; the wait is

$$g(x) = 15 - x, \quad 30 - x, \quad 45 - x, \quad 60 - x \qquad \text{on } (0, 15], \ (15, 30], \ (30, 45], \ (45, 60].$$

By Theorem 3-5,

$$\begin{aligned} E[g(X)] &= \frac{1}{60}\Big\{ \int_0^{15} (15 - x)\, dx + \int_{15}^{30} (30 - x)\, dx + \int_{30}^{45} (45 - x)\, dx + \int_{45}^{60} (60 - x)\, dx \Big\} \\ &= \frac{1}{60} \times 4 \times 112.5 = 7.5, \end{aligned}$$

i.e. 7 minutes 30 seconds.

Theorem 3-6 (Properties of expected value)

(1) $E(aX + b) = aE(X) + b$ (a, b constants).
(2) $E[c_1 g_1(X) + c_2 g_2(X)] = c_1 E[g_1(X)] + c_2 E[g_2(X)]$ (c_1, c_2 constants, g_1, g_2 any functions).

Proof. [Supplement] (2) By Theorem 3-5, in the discrete case

$$E[c_1 g_1(X) + c_2 g_2(X)] = \sum_x \{c_1 g_1(x) + c_2 g_2(x)\} p(x) = c_1 \sum_x g_1(x) p(x) + c_2 \sum_x g_2(x) p(x),$$

likewise with integrals.

(1) Take $g_1(x) = x$, $g_2 = 1$, $c_1 = a$, $c_2 = b$; $E(1) = 1$. ■

Definition 3-8 (Variance and standard deviation of a random variable)

If X has mean μ and p.m.f. (or p.d.f.) $p(x)$,

(1) $\mathrm{Var}(X) = E[(X - \mu)^2] = \begin{cases} \sum_{\text{all } x} (x - \mu)^2 p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} (x - \mu)^2 p(x)\, dx & \text{(continuous)} \end{cases}$

(2) $\mathrm{sd}(X) = \sqrt{\mathrm{Var}(X)} = \sqrt{E[(X - \mu)^2]}$.

For a population model these are σ^2 and σ.

Theorem 3-7 (Shortcut formula for the variance)

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2.$$

Proof. By Theorem 3-6(2) with $\mu = E(X)$,

$$\mathrm{Var}(X) = E[(X - \mu)^2] = E(X^2 - 2\mu X + \mu^2) = E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - \mu^2.$$

■

Example 3-12 (Two coin tosses)

X is the number of heads in 2 tosses of a coin. Find the mean and standard deviation of X.
Solution.

x	0	1	2	Total
$p(x)$	1/4	1/2	1/4	1
$xp(x)$	0	1/2	1/2	1
$x^2p(x)$	0	1/2	1	3/2

So $E(X) = 1$, $E(X^2) = \frac{3}{2}$, $\mathrm{Var}(X) = \frac{3}{2} - 1^2 = \frac{1}{2}$ and $\mathrm{sd}(X) = 1/\sqrt{2}$.

Example 3-13 (Spinner)

For the spinner of Illustration 3-4, find the mean and standard deviation of X.
Solution. $p(x) = 1$ on $[0, 1]$, so

$$E(X) = \int_0^1 x\,dx = \frac{1}{2}, \qquad E(X^2) = \int_0^1 x^2\,dx = \frac{1}{3},$$

$$\mathrm{Var}(X) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}, \qquad \mathrm{sd}(X) = \frac{1}{\sqrt{12}}.$$

Theorem 3-8 (Property of the variance)

$$\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X) \qquad (a, b \text{ constants}).$$

Proof. [Supplement] Let $\mu = E(X)$. By Theorem 3-6(1), $E(aX+b) = a\mu+b$, so $(aX+b) - E(aX+b) = a(X-\mu)$ and

$$\mathrm{Var}(aX + b) = E[a^2(X-\mu)^2] = a^2E[(X-\mu)^2] = a^2\,\mathrm{Var}(X).$$

■

Subtracting the mean and dividing by the standard deviation is **standardization**:

$$Z = \frac{X - E(X)}{\mathrm{sd}(X)}.$$

By Theorems 3-6 and 3-8 (with $a = 1/\mathrm{sd}(X)$, $b = -E(X)/\mathrm{sd}(X)$), $E(Z) = 0$ and $\mathrm{Var}(Z) = \mathrm{Var}(X)/\mathrm{sd}(X)^2 = 1$.

3.5 Joint distribution of two random variables

Motivation. Related variables (height and weight) need a joint distribution; we treat the discrete case.

Illustration 3-5: same marginals, different relations

Toss coins A, B, C: $X = 1$ if A is heads; $Y = 1$ if A, B are heads; $Z = 1$ if B, C are heads (else 0). Y and Z both have $P(0) = \frac{3}{4}$, $P(1) = \frac{1}{4}$, but $X = 0$ forces $Y = 0$, whereas given $X = 0$, Z is 0 or 1 with probabilities $\frac{3}{4}, \frac{1}{4}$.

Definition 3-9 (Joint distribution of two random variables)

The description of how probability is spread over all ordered pairs of values of X and Y, by nonnegative numbers summing to 1, is the **joint probability distribution** (or **joint distribution**) of X and Y.

For discrete X, Y with values $x_1, x_2, \dots$ and $y_1, y_2, \dots$,

$$p(x,y) = \begin{cases} P(X = x,\ Y = y), & (x,y) = (x_i, y_j)\ (i,j = 1,2,\dots) \\ 0, & \text{otherwise} \end{cases}$$

is the **joint probability mass function**.

Definition 3-10 (Joint probability mass function)

$p(x,y)$ is the **joint probability mass function** of the discrete random variables X and Y if
(1) $0 \le p(x,y) \le 1$ and $\sum_{\text{all } x}\sum_{\text{all } y} p(x,y) = 1$;
(2) $P(a < X \le b,\ c < Y \le d) = \sum_{a<x\le b}\sum_{c<y\le d} p(x,y)$.

Example 3-14 (Joint distributions from three coins)

For the experiment of Illustration 3-5, find the joint distribution of
(a) X and Y; (b) X and Z.
Solution. Listing coins A, B, C:

Outcome	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
X	1	1	1	1	0	0	0	0
Y	1	1	0	0	0	0	0	0
Z	1	0	0	0	1	0	0	0

Each outcome has probability $\frac{1}{8}$; collecting them,

(a) X and Y

$y\backslash x$	0	1	Row sum
0	2/4	1/4	3/4
1	0	1/4	1/4
Col. sum	2/4	2/4	1

(b) X and Z

$z\backslash x$	0	1	Row sum
0	3/8	3/8	6/8
1	1/8	1/8	2/8
Col. sum	4/8	4/8	1

Row and column sums give the **marginal probability mass functions**

$$p_1(x) = \sum_{\text{all } y} p(x,y), \qquad p_2(y) = \sum_{\text{all } x} p(x,y),$$

since $\{X = x\} = \bigsqcup_y \{X = x, Y = y\}$.

Example 3-15 (Two games on one die)

A game pays according to one die roll. A receives 100 won for 1 or 2, 200 won for 3 or 4, 300 won for 5 or 6; B receives 100 won for an even number and (face $\times 100$) won for an odd number. Find the distribution of Z, the total received by A and B.
Solution. With X, Y the amounts of A, B (e.g. $(100, 100)$ for faces 1, 2):

$x\backslash y$	100	300	500	Row sum
100	2/6	0	0	2/6
200	1/6	1/6	0	2/6
300	1/6	0	1/6	2/6
Col. sum	4/6	1/6	1/6	1

$Z = X + Y$ takes the values 200, 300, 400, 500, 800 with probabilities $\frac{2}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}$.

Theorem 3-9 (Expectation of a function of two random variables)

Let X, Y be discrete with joint p.m.f. $p(x, y)$. For any functions g, g_1, g_2:
(1) $E[g(X,Y)] = \sum_{\text{all } x} \sum_{\text{all } y} g(x,y)\, p(x,y)$;
(2) $E[c_1 g_1(X,Y) + c_2 g_2(X,Y)] = c_1 E[g_1(X,Y)] + c_2 E[g_2(X,Y)]$ (c_1, c_2 constants).

Proof. [Supplement] (1) For $W = g(X,Y)$, $p_W(w) = \sum_{(x,y):\, g(x,y)=w} p(x,y)$, so

$$E(W) = \sum_w w\, p_W(w) = \sum_w \sum_{(x,y):\, g(x,y)=w} g(x,y)\, p(x,y) = \sum_x \sum_y g(x,y)\, p(x,y),$$

(regrouping by absolute convergence).

(2) Apply (1) to $c_1 g_1 + c_2 g_2$ and split:

$$\sum_x \sum_y \{c_1 g_1 + c_2 g_2\}(x,y)\, p(x,y) = c_1 \sum_x \sum_y g_1(x,y) p(x,y) + c_2 \sum_x \sum_y g_2(x,y) p(x,y).$$

■

Definition 3-11 (Independence of two random variables)

Let $p(x,y)$ be the joint p.m.f. (joint p.d.f.) of discrete (continuous) X, Y and p_1, p_2 the marginals. X and Y are **independent** if for all x, y

$$p(x,y) = p_1(x)\, p_2(y).$$

Otherwise they are **dependent**.

Example 3-16 (Independence check)

A coin is tossed three times; X is the number of heads in the first two tosses and Y the number of heads on the third toss. Are X and Y independent?
Solution. The joint distribution is

$x\backslash y$	0	1	Row sum
0	1/8	1/8	1/4
1	2/8	2/8	1/2
2	1/8	1/8	1/4
Col. sum	1/2	1/2	1

Every cell is the product of its margins (e.g. $p_1(0)p_2(0) = \frac{1}{8} = p(0,0)$), so X and Y are independent.

3.6 Covariance and correlation

Motivation. The mean product of standardized deviations measures linear association.

Let X, Y have means μ_1, μ_2 and standard deviations σ_1, σ_2.

Definition 3-12 (Covariance and correlation coefficient)

(1) $\mathrm{Cov}(X,Y) = E[(X-\mu_1)(Y-\mu_2)]$.

(2) $\mathrm{Corr}(X,Y) = E\Big[\Big(\frac{X-\mu_1}{\sigma_1}\Big)\Big(\frac{Y-\mu_2}{\sigma_2}\Big)\Big] = \frac{\mathrm{Cov}(X,Y)}{\mathrm{sd}(X)\,\mathrm{sd}(Y)}$.

Proposition (Range of the correlation)

$-1 \le \mathrm{Corr}(X,Y) \le 1$; and if $Y = aX + b$ with $a \neq 0$, then $\mathrm{Corr}(X,Y) = 1$ for $a > 0$ and -1 for $a < 0$.

Proof. [Supplement]

Step 1. Let $U = (X-\mu_1)/\sigma_1$, $V = (Y-\mu_2)/\sigma_2$ and $\rho = \mathrm{Corr}(X,Y) = E(UV)$. Then $E(U^2) = E(V^2) = 1$, and by Theorem 3-9(2)

$$0 \le E[(U-\rho V)^2] = E(U^2) - 2\rho E(UV) + \rho^2 E(V^2) = 1 - \rho^2,$$

so $\rho^2 \le 1$.

Step 2. If $Y = aX + b$, then $\mu_2 = a\mu_1 + b$ and $\sigma_2 = |a|\sigma_1$ (Theorem 3-8), so $V = \frac{a}{|a|}U$ and $\rho = \frac{a}{|a|}E(U^2) = \pm 1$ according to the sign of a. ■

Theorem 3-10 (Shortcut formula for the covariance)

$$\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y).$$

Proof. By Theorem 3-9(2),

$$\begin{aligned}\mathrm{Cov}(X,Y) &= E[(X-\mu_1)(Y-\mu_2)] = E(XY - \mu_2 X - \mu_1 Y + \mu_1\mu_2)\\ &= E(XY) - \mu_2 E(X) - \mu_1 E(Y) + \mu_1\mu_2 = E(XY) - \mu_1\mu_2.\end{aligned}$$

■

Example 3-17 (Covariance and correlation)

X and Y have the joint distribution below. Find $\mathrm{Cov}(X,Y)$ and $\mathrm{Corr}(X,Y)$.

$x \backslash y$	0	1	2	3	Row sum
0	0.05	0.05	0.10	0.00	0.20
1	0.05	0.10	0.25	0.10	0.50
2	0.00	0.15	0.10	0.05	0.30
Col. sum	0.10	0.30	0.45	0.15	1.00

Solution. From the margins,

$$\mu_1 = 1.1, \qquad \sigma_1^2 = 1.7 - (1.1)^2 = 0.49, \qquad \sigma_1 = 0.7,$$

$$\mu_2 = 1.65, \qquad \sigma_2^2 = 3.45 - (1.65)^2 = 0.7275, \qquad \sigma_2 = \sqrt{0.7275} = 0.8529.$$

Summing $xy\,p(x,y)$ over the cells with $x, y, p \neq 0$,

$$E(XY) = 0.1 + 0.5 + 0.3 + 0.3 + 0.4 + 0.3 = 1.9.$$

Hence $\mathrm{Cov}(X,Y) = 1.90 - 1.65 \times 1.1 = 0.085$ and

$$\mathrm{Corr}(X,Y) = \frac{0.085}{0.70 \times 0.8529} = 0.1424.$$

Theorem 3-11 (Properties of covariance and correlation)

For constants a, b, c, d:
(1) $\mathrm{Cov}(aX+b,\, cY+d) = ac\ \mathrm{Cov}(X,Y)$;
(2) $\mathrm{Corr}(aX+b,\, cY+d) = \mathrm{Corr}(X,Y)$ if $ac > 0$, and $-\mathrm{Corr}(X,Y)$ if $ac < 0$.

Proof. [Supplement] (1) The centred variables are $a(X-\mu_1)$ and $c(Y-\mu_2)$, so

$$\mathrm{Cov}(aX+b, cY+d) = E[ac(X-\mu_1)(Y-\mu_2)] = ac\ \mathrm{Cov}(X,Y).$$

(2) By Theorem 3-8, $\mathrm{sd}(aX+b) = |a|\,\mathrm{sd}(X)$ and $\mathrm{sd}(cY+d) = |c|\,\mathrm{sd}(Y)$. With (1),

$$\mathrm{Corr}(aX+b, cY+d) = \frac{ac\ \mathrm{Cov}(X,Y)}{|a||c|\,\mathrm{sd}(X)\,\mathrm{sd}(Y)} = \frac{ac}{|ac|}\,\mathrm{Corr}(X,Y),$$

with $ac/|ac| = \pm 1$. ■

Theorem 3-12 (Variance of a sum of two random variables)

(1) $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$.
(2) $\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X,Y)$.

Proof. [Supplement] $(X \pm Y) - E(X \pm Y) = (X-\mu_1) \pm (Y-\mu_2)$; square and take expectations (Theorem 3-9(2)):

$$\mathrm{Var}(X \pm Y) = E[(X-\mu_1)^2] + E[(Y-\mu_2)^2] \pm 2E[(X-\mu_1)(Y-\mu_2)],$$

which is $\mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\,\mathrm{Cov}(X,Y)$. ■

Theorem 3-13 (Independent random variables)

If X and Y are independent, then
(1) $E(XY) = E(X)E(Y)$;
(2) $\mathrm{Cov}(X,Y) = 0$ and $\mathrm{Corr}(X,Y) = 0$;
(3) $\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.

Proof. (1) By Theorem 3-9(1) and $p(x,y) = p_1(x)p_2(y)$,

$$E(XY) = \sum_x \sum_y xy\,p(x,y) = \sum_x \sum_y xy\,p_1(x)p_2(y) = \sum_x x\,p_1(x) \sum_y y\,p_2(y) = E(X)E(Y).$$

(2) By Theorem 3-10, $\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y) = 0$, hence also $\mathrm{Corr}(X,Y) = 0$.

(3) Insert $\mathrm{Cov}(X,Y) = 0$ into Theorem 3-12. ■